

FACTORIAL PROGRAM

```
#include <stdio.h>

unsigned long long factorial(int n) {
    if (n == 0 || n == 1) {
        return 1;
    }
    return n * factorial(n - 1);
}

int main() {
    int num;
    printf("Enter a positive integer: ");
    scanf("%d", &num);

    if (num < 0) {
        printf("Error: Factorial is not defined for negative numbers.\n");
    } else {
        printf("Factorial of %d is %llu\n", num, factorial(num));
    }

    return 0;
}
```

Amstrong Numbers

```
#include <stdio.h>

int main() {
    int num, originalNum, remainder, result = 0;

    printf("Enter a 3-digit integer: ");
    scanf("%d", &num);

    originalNum = num;

    while (originalNum != 0) {
        remainder = originalNum % 10;
        result += remainder * remainder * remainder;
        originalNum /= 10;
    }

    if (result == num)
```

```

        printf("%d is an Armstrong number.\n", num);
    else
        printf("%d is not an Armstrong number.\n", num);

    return 0;
}

```

Arrange the names in alphabetical order.

```

#include <stdio.h>
#include <string.h>

#define MAX_NAMES 5
#define MAX_LENGTH 50

int main() {
    char names[MAX_NAMES][MAX_LENGTH];
    char temp[MAX_LENGTH];

    printf("Enter %d names:\n", MAX_NAMES);
    for (int i = 0; i < MAX_NAMES; i++) {
        scanf("%s", names[i]);
    }

    for (int i = 0; i < MAX_NAMES - 1; i++) {
        for (int j = i + 1; j < MAX_NAMES; j++) {
            if (strcmp(names[i], names[j]) > 0) {
                strcpy(temp, names[i]);
                strcpy(names[i], names[j]);
                strcpy(names[j], temp);
            }
        }
    }

    printf("\nNames in alphabetical order:\n");
    for (int i = 0; i < MAX_NAMES; i++) {
        printf("%s\n", names[i]);
    }

    return 0;
}

```

MATRIX ADDITION

```
#include <stdio.h>

#define ROWS 3
#define COLS 3

int main() {
    int A[ROWS][COLS] = {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}};
    int B[ROWS][COLS] = {{9, 8, 7}, {6, 5, 4}, {3, 2, 1}};
    int result[ROWS][COLS];

    // Print Matrix A
    printf("Matrix A:\n");
    for (int i = 0; i < ROWS; i++) {
        for (int j = 0; j < COLS; j++) {
            printf("%d ", A[i][j]);
        }
        printf("\n");
    }

    // Print Matrix B
    printf("\nMatrix B:\n");
    for (int i = 0; i < ROWS; i++) {
        for (int j = 0; j < COLS; j++) {
            printf("%d ", B[i][j]);
        }
        printf("\n");
    }

    // Matrix Addition
    printf("\nMatrix Addition (A + B):\n");
    for (int i = 0; i < ROWS; i++) {
        for (int j = 0; j < COLS; j++) {
            result[i][j] = A[i][j] + B[i][j];
            printf("%d ", result[i][j]);
        }
        printf("\n");
    }
}
```

MATRIX MULTIPLICATION

```
#include <stdio.h>
#define ROWS 3
#define COLS 3

int main() {
    int A[ROWS][COLS] = {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}};
    int B[ROWS][COLS] = {{9, 8, 7}, {6, 5, 4}, {3, 2, 1}};
    int result[ROWS][COLS];

    // Print Matrix A
    printf("Matrix A:\n");
    for (int i = 0; i < ROWS; i++) {
        for (int j = 0; j < COLS; j++) {
            printf("%d ", A[i][j]);
        }
        printf("\n");
    }

    // Print Matrix B
    printf("\nMatrix B:\n");
    for (int i = 0; i < ROWS; i++) {
        for (int j = 0; j < COLS; j++) {
            printf("%d ", B[i][j]);
        }
        printf("\n");
    }

    // Matrix Multiplication
    printf("\nMatrix Multiplication (A * B):\n");
    for (int i = 0; i < ROWS; i++) {
        for (int j = 0; j < COLS; j++) {
            result[i][j] = 0;
            for (int k = 0; k < COLS; k++) {
                result[i][j] += A[i][k] * B[k][j];
            }
            printf("%d ", result[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

C program to check if a number is a palindrome or not.

```
#include <stdio.h>

int main() {
    int num, originalNum, reversedNum = 0, remainder;

    printf("Enter an integer: ");
    scanf("%d", &num);

    originalNum = num;

    // Reverse the number
    while (num != 0) {
        remainder = num % 10;
        reversedNum = reversedNum * 10 + remainder;
        num /= 10;
    }

    // Check if palindrome
    if (originalNum == reversedNum) {
        printf("%d is a palindrome.\n", originalNum);
    } else {
        printf("%d is not a palindrome.\n", originalNum);
    }

    return 0;
}
```

Program to check if a number is prime:

```
#include <stdio.h>

int main() {
    int n, isPrime = 1;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    if (n <= 1) isPrime = 0;

    for (int i = 2; i * i <= n; i++) {
        if (n % i == 0) {
```

```

        isPrime = 0;
        break;
    }
}

if (isPrime)
    printf("%d is prime.\n", n);
else
    printf("%d is not prime.\n", n);

return 0;
}

```

Program to print Fibonacci series:

```

#include <stdio.h>

int main() {
    int n, first = 0, second = 1, next;

    printf("Enter the number of terms: ");
    scanf("%d", &n);

    printf("Fibonacci Series: ");

    for (int i = 0; i < n; i++) {
        if (i <= 1)
            next = i;
        else {
            next = first + second;
            first = second;
            second = next;
        }
        printf("%d ", next);
    }

    return 0;
}

```

Program to reverse a number:

```

#include <stdio.h>

```

```

int main() {
    int num, reversed = 0, remainder;

    printf("Enter an integer: ");
    scanf("%d", &num);

    while (num != 0) {
        remainder = num % 10;
        reversed = reversed * 10 + remainder;
        num /= 10;
    }

    printf("Reversed number: %d\n", reversed);
    return 0;
}

```

Program to check if a year is a leap year:

```

#include <stdio.h>

int main() {
    int year;

    printf("Enter a year: ");
    scanf("%d", &year);

    if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0))
        printf("%d is a leap year.\n", year);
    else
        printf("%d is not a leap year.\n", year);

    return 0;
}

```

Program to swap two numbers:

```

#include <stdio.h>

int main() {
    int a, b, temp;

    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);

    printf("Before swap: a = %d, b = %d\n", a, b);
}

```

```

temp = a;
a = b;
b = temp;

printf("After swap: a = %d, b = %d\n", a, b);
return 0;
}

```

Program to find the sum of digits of a number:

```

#include <stdio.h>

int main() {
    int num, sum = 0, digit;

    printf("Enter a number: ");
    scanf("%d", &num);

    while (num > 0) {
        digit = num % 10;
        sum += digit;
        num /= 10;
    }

    printf("Sum of digits: %d\n", sum);
    return 0;
}

```

Program to check if a string is a palindrome:

```

#include <stdio.h>
#include <string.h>

int main() {
    char str[100];
    int i, length;
    int flag = 0;

    printf("Enter a string: ");
    scanf("%s", str);

    length = strlen(str);

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for(i = 0; i < length/2; i++) {
    if(str[i] != str[length-1-i]) {
        flag = 1;
        break;
    }
}

if (flag) {
    printf("%s is not a palindrome", str);
}
else {
    printf("%s is a palindrome", str);
}

return 0;
}

```

Program to find the transpose of a matrix:

```

#include <stdio.h>

int main() {
    int a[10][10], transpose[10][10], r, c, i, j;

    printf("Enter rows and columns: ");
    scanf("%d %d", &r, &c);

    printf("\nEnter matrix elements:\n");
    for(i = 0; i < r; ++i)
        for(j = 0; j < c; ++j) {
            printf("Enter element a%d%d: ", i + 1, j + 1);
            scanf("%d", &a[i][j]);
        }

    printf("\nEnter matrix: \n");
    for(i = 0; i < r; ++i)
        for(j = 0; j < c; ++j) {
            printf("%d ", a[i][j]);
            if (j == c - 1)
                printf("\n");
        }

    for(i = 0; i < r; ++i)
        for(j = 0; j < c; ++j) {

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        transpose[j][i] = a[i][j];
    }

    printf("\nTranspose of the matrix:\n");
    for(i = 0; i < c; ++i)
        for(j = 0; j < r; ++j) {
            printf("%d ", transpose[i][j]);
            if (j == r - 1)
                printf("\n");
        }

    return 0;
}
```